

CAR PRICE PREDICTION USING MACHINE LEARNING

Data Science & Analytics Internship

CodeBase Studio

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1. Objective

The objective of this project is to develop a machine learning model that predicts car selling prices based on vehicle characteristics such as year, present price, kilometers driven, fuel type, seller type, transmission type, and ownership details.

2. Dataset Description

The dataset contains information about used cars and their selling prices.

Total Records: 301

Total Variables: 9

Variables:

- Car_Name
- Year
- Selling_Price
- Present_Price
- Driven_kms
- Fuel_Type
- Selling_type
- Transmission
- Owner

The target variable is Selling_Price, which represents the selling price of a car.

3. Dataset Preview

3. Load dataset

df = pd.read_csv('car_data (1).csv')

df.head()

...

	Car_Name	Year	Selling_Price	Present_Price	Driven_kms	Fuel_Type	Selling_type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0

Next steps:

Generate code with df

New interactive sheet

Explanation

The first five records of the dataset were displayed to understand the available features and target variable. The dataset contains information related to car specifications and selling prices.

4. Dataset Information

4. Check dataset info

df.info()

```

.. <class 'pandas.core.frame.DataFrame'>
RangeIndex: 301 entries, 0 to 300
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Car_Name        301 non-null   object
1   Year            301 non-null   int64
2   Selling_Price   301 non-null   float64
3   Present_Price   301 non-null   float64
4   Driven_kms      301 non-null   int64
5   Fuel_Type       301 non-null   object
6   Selling_type    301 non-null   object
7   Transmission    301 non-null   object
8   Owner           301 non-null   int64
dtypes: float64(2), int64(3), object(4)
memory usage: 21.3+ KB

```

Explanation

The dataset contains 301 records and 9 columns. It includes numerical and categorical variables. Numerical variables include Year, Selling_Price, Present_Price, Driven_kms, and Owner, while Fuel_Type, Selling_type, Transmission, and Car_Name are categorical variables.

5. Missing Value Analysis

```
# 5. Check missing values
df.isnull().sum()

...
0
Car_Name    0
Year        0
Selling_Price  0
Present_Price  0
Driven_kms   0
Fuel_Type    0
Selling_type  0
Transmission  0
Owner        0
dtype: int64
```

Explanation

The missing value analysis indicates that all columns contain zero missing values. Therefore, the dataset is complete and suitable for machine learning analysis without additional data cleaning.

6. Data Preprocessing

The categorical variables were converted into numerical values using Label Encoding. This transformation allowed the machine learning model to process the data effectively.

The dataset was then divided into:

- Features (X)
- Target Variable (y)

The data was split into training and testing sets using an 80:20 ratio.

7. Model Development

A Linear Regression algorithm was used to build the prediction model.

Steps performed:

1. Data Loading
2. Data Preprocessing
3. Feature Encoding
4. Train-Test Split
5. Model Training
6. Prediction
7. Performance Evaluation

8. Model Evaluation

```
+from sklearn.metrics import mean_absolute_error, mean_squared_error
+
# 12. Model evaluation
print("MAE:", mean_absolute_error(y_test, predictions))
print("MSE:", mean_squared_error(y_test, predictions))
print("R2 Score:", r2_score(y_test, predictions))
```

```
MAE: 1.2147329896540084
MSE: 3.537020423742584
R2 Score: 0.8464540623821766
```

Results:

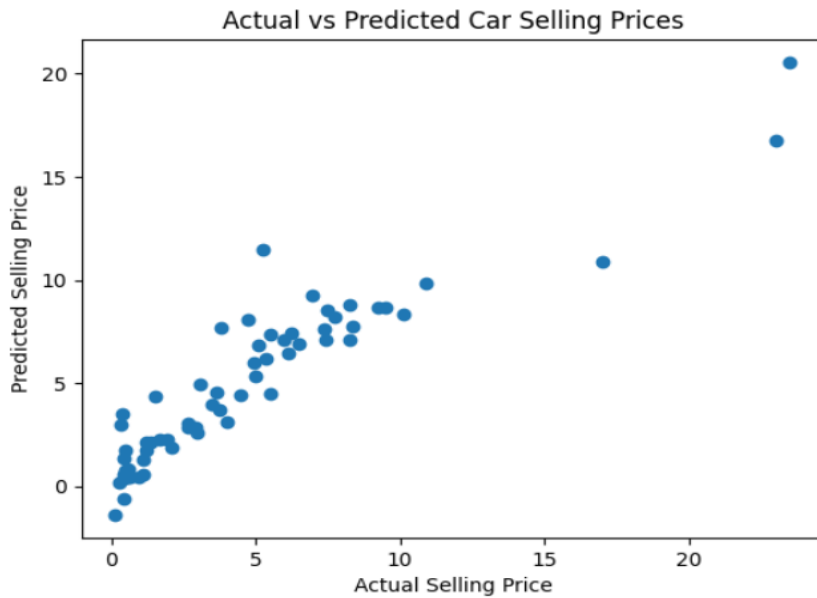
- Mean Absolute Error (MAE): 1.2147
- Mean Squared Error (MSE): 3.5370
- R² Score: 0.8465

Interpretation

The R² score of 0.8465 indicates that approximately 84.65% of the variation in car selling prices is explained by the model. This demonstrates strong predictive performance.

The low MAE value indicates that the prediction error is relatively small.

9. Actual vs Predicted Price Analysis



Explanation

The scatter plot compares actual selling prices with predicted selling prices.

Most data points are located close to the diagonal trend, indicating that the model's predictions are reasonably accurate. The graph demonstrates a strong positive relationship between actual and predicted values.

10. Findings

1. The dataset contains no missing values.
2. Present price and vehicle characteristics significantly influence selling price.
3. The Linear Regression model achieved strong predictive performance.
4. The model successfully predicts car selling prices using historical vehicle data.
5. The Actual vs Predicted graph confirms the effectiveness of the model.

11. Real-World Applications

Car price prediction models can be used by:

- Car dealerships
- Online vehicle marketplaces
- Insurance companies
- Automobile valuation services

These models help estimate fair vehicle prices and support business decision-making.

12. Conclusion

A machine learning model was successfully developed to predict car selling prices using Linear Regression. The model achieved an R^2 score of 0.8465, indicating strong predictive capability. The results demonstrate how machine learning techniques can be applied to solve real-world pricing problems in the automobile industry.